

Assignment Day 5 - Arrays

1. Maximum Element

```
#include <iostream>
using namespace std;

int main() {
    int arr[5] = {10, 25, 8, 40, 15};
    int max = arr[0];

    for(int i = 1; i < 5; i++) {
        if(arr[i] > max)
            max = arr[i];
    }

    cout << "Maximum Element = " << max;
    return 0;
}
```

Input: No Input

Output: Maximum Element = 40

2. Minimum Element

```
#include <iostream>
using namespace std;

int main() {
    int arr[5] = {10, 25, 8, 40, 15};
    int min = arr[0];

    for(int i = 1; i < 5; i++) {
        if(arr[i] < min)
            min = arr[i];
    }

    cout << "Minimum Element = " << min;
    return 0;
}
```

Input: No Input

Output: Minimum Element = 8

3. Array Sum

```
#include <iostream>
using namespace std;

int main() {
    int arr[5] = {10, 20, 30, 40, 50};
    int sum = 0;

    for(int i = 0; i < 5; i++)
        sum += arr[i];

    cout << "Sum = " << sum;
    return 0;
}
```

Input: No Input

Output: Sum = 150

4. Reverse Array

```
#include <iostream>
using namespace std;

int main() {
    int arr[5] = {1, 2, 3, 4, 5};

    for(int i = 4; i >= 0; i--)
        cout << arr[i] << " ";

    return 0;
}
```

Input: No Input

Output: 5 4 3 2 1

5. Second Largest Element

```
#include <iostream>
using namespace std;

int main() {
    int arr[5] = {10, 25, 8, 40, 15};
    int largest = arr[0], second = -1;

    for(int i = 0; i < 5; i++) {
        if(arr[i] > largest) {
            second = largest;
            largest = arr[i];
        }
        else if(arr[i] > second && arr[i] != largest) {
            second = arr[i];
        }
    }

    cout << "Second Largest = " << second;
    return 0;
}
```

Input: No Input

Output: Second Largest = 25

6. Matrix Addition

```
#include <iostream>
using namespace std;

int main() {
    int A[2][2] = {{1,2},{3,4}};
    int B[2][2] = {{5,6},{7,8}};
    int C[2][2];

    for(int i=0;i<2;i++) {
        for(int j=0;j<2;j++) {
            C[i][j] = A[i][j] + B[i][j];
        }
    }

    for(int i=0;i<2;i++) {
        for(int j=0;j<2;j++) {
            cout << C[i][j] << " ";
        }
        cout << endl;
    }

    return 0;
}
```

Input: No Input

Output: 6 8 10 12